Skills Summary

Factoring Special Forms: Squares

Use this reference while completing your worksheet or when studying.

Difference of squares: $A^2 - B^2 = (A-B)(A+B)$ — two terms, subtraction, both perfect squares.

Perfect-square trinomial: $A^2 + 2AB + B^2 = (A+B)^2$ and $A^2 - 2AB + B^2 = (A-B)^2$ — three terms, ends both squares, middle equal to twice the product of their roots.

No pattern for a sum of squares: $A^2 + B^2$ does not factor over the integers. Recognising that saves time on every “factor completely”.

Squares worth knowing on sight: $4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144$; and $4x^2 = (2x)^2$, $9x^2 = (3x)^2$, $x^4 = (x^2)^2$.

1. Difference of two squares

Two terms and a minus sign is the trigger. Name A and B as the square roots of the two terms, then write $(A-B)(A+B)$. The check is quick: expanding always cancels the middle terms.

Example: Factor $x^2 - 36$.
Step 1. Two terms, subtraction, and both terms are perfect squares — that is exactly the difference-of-squares trigger, so no trial and error is needed.
Step 2. $A = x$ because $x^2 = (x)^2$, and $B = 6$ because $36 = (6)^2$. $x^2 - 36 = \boldsymbol{(x-6)(x+6)}$

Try it: Factor $x^2 - 121$.

check: $\boldsymbol{(x-11)(x+11)}$

2. Perfect-square trinomial

Three terms with square ends. Take the roots of the first and last terms, then *test* the middle: it must equal $2AB$. If it does not, the trinomial is not a perfect square and needs ordinary factoring instead. The sign of the middle term becomes the sign inside the bracket.

Example: Factor $x^2 + 6x + 9$.
Step 1. Three terms whose first and last are squares — test the perfect-square pattern before trying factor pairs.
Step 2. $A = x$, $B = 3$, and the test is $2AB = 2 \cdot x \cdot 3 = 6x$, which matches the middle term. The middle sign is positive, so the bracket holds a sum. $x^2 + 6x + 9 = \boldsymbol{(x+3)^2}$

Try it: Factor $x^2 - 8x + 16$.

check: $\boldsymbol{(x-4)^2}$

3. Common factor first, then the pattern

Before testing either pattern, ask whether every term shares a factor. A coefficient that is not a perfect square ($2x^2$, $3x^2$, $18x^2$) is the usual sign that a common factor is hiding the pattern. Take it out, factor what is left, and keep the common factor in the final answer.

Example: Factor $3x^2 - 27$ completely.
Step 1. $3x^2$ is not a perfect square, so the difference-of-squares pattern cannot apply yet — but both terms are divisible by 3, so remove that first.
Step 2. $3x^2 - 27 = 3(x^2 - 9)$, and $x^2 - 9 = (x-3)(x+3)$. $3x^2 - 27 = \mathbf{3(x-3)(x+3)}$

Try it: Factor $4x^2 - 100$ completely.

check: $\mathbf{4(x-5)(x+5)}$

(!) Watch out: $x^2 - 9$ is *not* $(x-3)^2$. Squaring a binomial always creates a middle term, so $(x-3)^2 = x^2 - 6x + 9$. Count the terms first: two terms means difference of squares, three terms means look for a perfect square. And "factor completely" means keep going — $x^4 - 81$ splits twice.